When Can Machine-Labelled Public Opinion Be Compared? Measurement Equivalence Across Platforms, Languages and LLM Annotators in Asian Social Media

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1. Problem

Communication research now routinely measures public opinion by having machines read social media. Sentiment, stance, toxicity and narrative frames are labelled by fine-tuned classifiers or, increasingly, by large language models (Gilardi, Alizadeh and Kubli, 2023), and the resulting scores are pooled across platforms, tracked over time and compared across languages. Each comparison quietly assumes that the instrument measures the same thing everywhere it reads. That assumption is rarely tested. Platforms differ in who posts and how they write (Jaidka, Guntuku and Ungar, 2018), and models can produce labels that are accurate on the surface yet miss what makes content socially consequential, the "telic errors" that Jaidka and Huang identify in vision-language annotators of harmful content. If a "negative" post means something different on TikTok than on Facebook, or in Singlish than in standard English, then differences between platforms or groups may be artefacts of measurement rather than findings about society.

2. Prior work and my starting point

With Dr Ella Haig I recently set out a protocol for this problem (Ikram and Haig, 2026, arXiv:2610.16757). It treats each platform as a survey mode and borrows tools from survey methodology: anchor events fixed in advance, a generalizability study against human labels, tests of measurement invariance and differential item functioning (DIF), and score linking onto a common scale. In a known-truth simulation with 500 replications, pooling platform scores at face value reported the wrong direction of daily sentiment on 13.8 of 60 days on average, against 0.8 days for an adjusted index. The paper came out of my own MSc dissertation (255,446 comments on UK economic policy across X, Reddit, YouTube and Quora), where I found that my BERT classifier's 88 per cent accuracy only measured agreement with the lexicon that produced its training labels. Two gaps remain, and both matter especially for Asia: the protocol has not been run on real data, and it has not been extended to languages, code-mixing and changing LLM annotators.

3. Research questions

1. **Platforms and languages.** When are LLM-labelled constructs (sentiment, stance, incivility, narrative frames) measurement-equivalent across platforms and across languages and code-mixed registers, such as English, Singlish, Malay, Mandarin and Urdu?
2. **Annotators over time.** How far do labels shift when the annotating model or prompt changes, and how can a measurement series be kept comparable across model versions?
3. **Consequences.** When the measures are corrected, do substantive conclusions about public reaction, polarisation or the spread of narratives across platforms change?

4. Data and methods

Study 1 (simulation, year 1). Extend the existing simulation so that platform, language, annotator model, prompt and time are all facets, and estimate how many human-labelled posts, and which anchor events, are needed to detect a difference large enough to reverse a conclusion. This needs no new data and sets the design for the empirical studies.

Study 2 (empirical, years 2 to 3). Collect public posts around a set of policy events fixed in advance (for example, cost-of-living measures, elections or public-health announcements in Singapore and in Pakistan) from several platforms through official APIs and permitted research access, following NUS ethics procedures. Label every post with several LLMs and fine-tuned models, and draw a stratified sample per platform and language for blind labelling by trained native-speaker coders, with a second coder on a subset. Run a G-study to split variance into platform, language, model, prompt and time; test invariance and DIF for each construct; and use

linear concept erasure (Fan et al., 2025) to check whether removing the platform signal also removes real differences. Correct downstream estimates with the human sample (Egami et al., 2023).

Study 3 (substantive, years 3 to 4). Re-run a substantive analysis, such as how a narrative travels between platforms or how polarised reactions to a policy appear, with corrected and uncorrected measures, and report when the conclusion changes. Network methods (community detection and diffusion) will be used to trace cross-platform spread.

5. Why NUS CNM, and feasibility

CNM combines computational communication with a focus on Asian media systems, and Associate Professor Kokil Jaidka's work on computational measurement of social media language, cross-platform differences and LLM annotation is the closest fit to this project; Dr Subhayan Mukerjee's work on news audiences across platforms would inform Study 3. Singapore's multilingual, code-mixed online public is an ideal setting to test whether measurement travels across languages. On feasibility: I have collected and labelled social media data at scale for nearly eight years, at Dubizzle Labs (data in English, Arabic, Urdu and Hindi across nine countries), at M+C Saatchi Fluency (LLM tagging with prompt-version tracking for global brands) and as CTO of Artemis AI, where our own evaluation showed that agreement between models is not accuracy. The simulation code is public, and the statistical methods are established; what is new is applying them with platforms, languages and models as the groups.

6. Expected contribution

A tested protocol and open toolkit for checking when machine-labelled public opinion can be compared; evidence on how much platform, language and model choice distort findings in Asian social media; and practical guidance for researchers, newsrooms and public bodies that rely on social listening.

Selected references

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